

Real-Time Flow State Analysis in Game: A Non-Intrusive System Based on Eye Feature Detection

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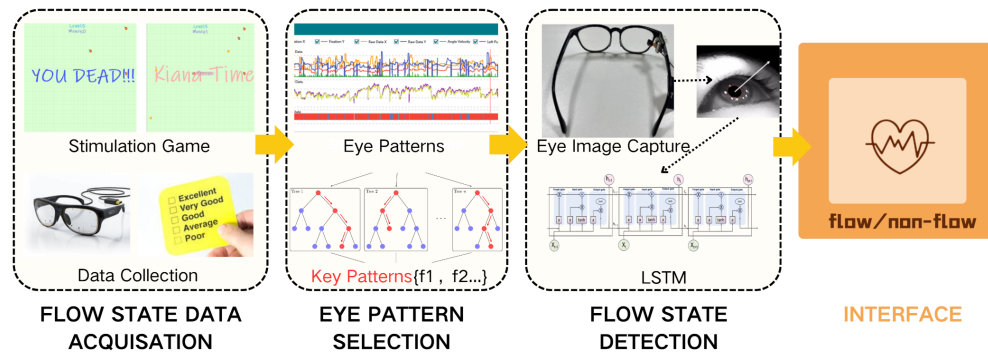


Figure 1: Pipeline of flow state detection system manipulation expense

ABSTRACT

Flow is a key indicator for deep immersive state. Current methods for detecting flow often rely on intrusive physiological devices (e.g EEG, HRV), requiring specialized equipment and complex data processing, with no capability for real-time flow state visualization. These limitations hinder the possibility to analyze the relationship between game settings and flow state fluctuations. To address this, we utilized an eye-tracking device (Tobii glasses) to record eye movement features in games under different flow states, creating a dataset to identify significant eye movement patterns correlated with flow state. Based on this, we developed a non-intrusive eye-tracking system that captures and analyzes specific eye features through a custom-designed algorithm, providing real-time feedback on flow states. Our system achieved a flow state detection accuracy of 85.3%, with a 412 ms detection delay, outperforming existing methods. This innovation offers a powerful tool for exploring the relationship between flow and game design, and further unlocking the biophysical mechanisms of flow.

Index Terms: Flow, Eye feature, Game, Immersion, Non-intrusive detection.

1 INTRODUCTION

Flow is an important indicator of user immersion [1], but current methods like questionnaires can't monitor flow in real-time, and physiological signal-based detection is intrusive and may interfere with the flow state. Eye tracking is simple, non-intrusive, and research shows eye movements are closely linked to flow[2].

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To address this, we propose a non-contact flow state detection system based on eye movement patterns. We initially selected the most significant eye movement patterns that correlated with flow states. Subsequently, we developed a wearable device that captures eye movement data and incorporated a trained LSTM model for classifying flow states (flow versus non-flow). An interface to output the flow state corresponding to the game scene is developed.

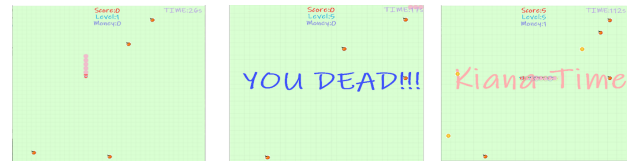


Figure 2: The screenshots of the test game

1.1 Dataset Collection and Annotation

This study used the Tobii Glasses 3 eye tracker to record eye movement data from 95 participants (university teachers and students, aged 18–35) during gameplay. IRB approval was obtained from the university's Ethics Committee. To calibrate difficulty, we developed a customized version of the classic Snake game (Figure 2), with participants using arrow keys to control the snake. The data acquisition phase began with a 4-minute resting period to record baseline physiological parameters. During gameplay, participants randomly selected one of three modes for 5 minutes. The simple, repetitive nature of Snake ensured a consistent gaming experience. After each session, participants completed a Flow Experience Assessment Scale (FEAS). Figure 3 illustrates the quiet, relaxed experimental environment in which participants were tested under optimal physical and emotional conditions.

1.2 Eye Pattern Extraction

We collected a comprehensive eye movement dataset, including 2D/3D gaze coordinates, gaze direction, pupil data, eye move-

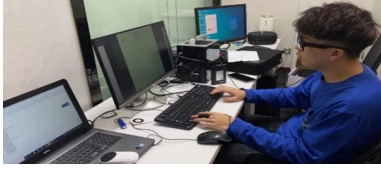


Figure 3: The experimental venue

ment types, fixation points, and sensor readings (magnetometer, gyroscope, accelerometer, blink frequency, and subjective labels). Invalid samples were removed, resulting in 95 high-quality datasets. From these, 34 key eye movement features were extracted. Using Random Forest (RF), we identified four critical features—saccadic velocity, pupil diameter, blink frequency, and fixation points—strongly associated with flow state. These features formed the basis of our real-time flow state detection system, with Figures 4 showing that flow states correspond to reduced blink frequency and enlarged pupil size.

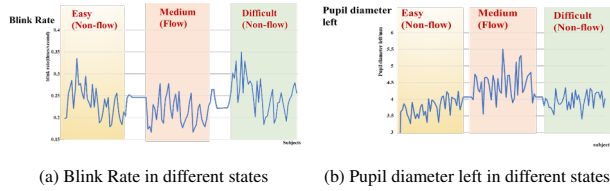


Figure 4: Blink rate and pupil diameter in different states

1.3 Flow State Detection

We aim to develop a real-time flow state detection system using eye movement data. A wearable device captures eye images, which are then transmitted to a server for flow state analysis. Reflection-based gaze tracking is employed for image capture, with the OV2640 camera used for imaging and the ESP32-S3 microcontroller handling data processing. This setup ensures real-time, low-latency transmission of video frames for accurate flow detection.(Figure 5).

We implemented a LSTM neural network architecture to detect the flow state by analyzing the eye movement data. The output from the final time step is passed through a fully connected layer to produce binary class scores for flow and non-flow states. The extracted eye patterns in previous sections is imputed into the LSTM network for flow state detection.

To train the LSTM model, we employed the binary cross-entropy loss function, a standard choice for classification tasks. The loss function L is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where y_i represents the true label for the i -th sample, $\hat{y}_i$ is the predicted probability, and N is the total number of samples.

2 RESULTS AND PERFORMANCE EVALUATION

We developed a visualization interface integrating the algorithm and display components. As depicted in Figure 6, the interface visualization consists of three integrated components and allows both high-level and granular interpretation of the system's flow state detection performance.

The analysis showed a significant positive correlation between the system's predictions and participants' subjective flow reports

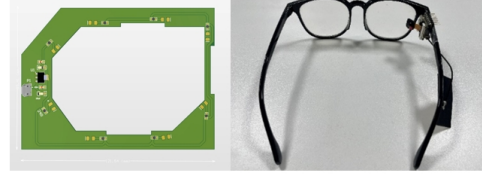


Figure 5: Eye movement acquisition device

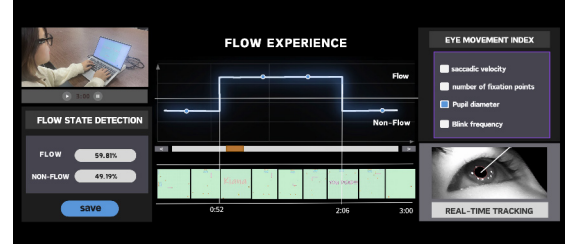


Figure 6: Real-time flow state detection and analysis interface

(PLCC = 0.76), indicating strong alignment with their actual experiences. The LSTM model achieved an accuracy of 85.3%. Compared to other models using multiple physiological signals [3][4], our unimodal LSTM model demonstrated similar or improved performance.

By marking the stimulus onset with a time-stamped signal and correlating it with the recorded eye-tracking data, we ascertained the delay between the actual pupil response and its registration by the device. Across several iterations and subsequent averaging of the data, we established that the eye-tracking system exhibits a latency of approximately 412 milliseconds.

3 CONCLUSION AND FUTURE WORK

This paper introduces a novel eye movement-based system for real-time game flow state detection, while current immersive devices typically feature eye-tracking functionality, they are limited to extracting only the gaze region, with basic gaze vectors remaining inaccessible. Our work overcomes this device limitation by enabling the extraction of comprehensive gaze vectors, providing a more detailed and accurate evaluation of immersion. We plan to integrate the Raspberry Pi AI to enable the system to function as an independent eye-tracking device in the future. This upgrade will enhance real-time performance and reduce hardware dependency. Additionally, we will explore the use of modern architectures, such as Transformers, to further improve the system's efficiency and accuracy.

ACKNOWLEDGMENTS

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